



CALICUT UNIVERSITY – FOUR-YEAR UNDER GRADUATE PROGRAMME (CU-FYUGP)

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	INTRODUCTORY QUANTUM FIELD THEORY				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	PHY5CJ303- Quantum Mechanics I and PHY7CJ403- Quantum Mechanics II				
Course Summary	The course provides a comprehensive overview of classical field theory, followed by a detailed exploration of the quantization processes for scalar fields, Dirac fields, and the electromagnetic field, aiming to elucidate the fundamental principles underlying modern theoretical physics				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Demonstrate an understanding of classical field theory and its applications in describing physical phenomena	Understand	Conceptual Understanding	Examinations, Assignments
CO2	Apply quantization techniques to scalar fields, Dirac fields, and the electromagnetic field	Apply	Application	Problem Sets, Lab Reports

CO3	Analyze the consequences of field quantization on particle interactions and quantum field theory	Analyze	Application	Research Papers, Projects
CO4	Evaluate the mathematical formalism of field quantization and its consistency with experimental observations	Evaluate	Conceptual Understanding	Presentations, Discussions
CO5	Explain the implications of field quantization for relativistic quantum mechanics and gauge theories	Understand	Conceptual Understanding	Written Reports, Essays
CO6	Synthesize knowledge of classical field theory and field quantization to propose solutions to theoretical problems in modern physics	Create	Synthesis	Capstone Projects, Oral Defenses
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F), Conceptual Knowledge (C), Procedural Knowledge (P), Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	CLASSICAL FIELD THEORY		10	15
	1	Why Quantum Field Theory, Creation and annihilation operators	2	
	2	Special relativity, Space and time in relativistic quantum theory	2	
	3	A quick review of particle mechanics, Euler-Lagrange equations in field theory	2	
	4	Hamiltonian formalism	2	
	5	Noether's theorem	2	
	Sections 1.1 – 1.4, 2.1 – 2.4 of Book 1			
II	QUANTIZATION OF SCALAR FIELDS		10	18
	6	Equation of motion	2	
	7	The field and its canonical quantization	2	
	8	Fourier decomposition of the field	2	
	9	Ground state of the Hamiltonian and normal ordering	3	

	10	Fock space	1	
	Sections 3.1 – 3.5 of Book 1			
III	QUANTIZATION OF DIRAC FIELDS		17	20
	11	Dirac Hamiltonian	2	
	12	Dirac equation	2	
	13	Plane wave solutions of Dirac equation	2	
	14	Projection operators	4	
	15	Lagrangian for a Dirac field .	2	
	16	Fourier decomposition of the field	3	
	17	Propagator	2	
	Sections 4.1 – 4.7 of Book 1			
IV	QUANTIZATION OF THE ELECTROMAGNETIC FIELD		11	17
	18	Problems with quantization	3	
	19	Modifying the classical Lagrangian	2	
	20	Propagator	2	
	21	Fourier decomposition of the field	2	
	22	Physical states	2	
	Sections 8.2 – 8.6 of Book 1			
V	OPEN ENDED MODULE			
	QUANTUM ELECTRODYNAMICS		12	

Books and References:

1. A First Book of Quantum Field Theory by Amitabha Lahiri and Palash B Pal, 2nd Edn(Book 1)
2. Quantum Field theory, Lewis H. Ryder, (Cambridge University Press -1995)
3. Field Theory – A modern primer – Pierre Ramond (Bengamin – 1996)
4. Quantum Field theory, Itzyskon and Zuber (McGraw Hill – 1989)
5. Quantum Field theory, Karson Huang (Wiley)

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	0	0	0	0	0	2	2	0	0	0	0	1
CO 2	2	1	0	0	0	0	2	2	0	0	0	0	1
CO 3	2	2	1	0	0	0	2	2	0	0	0	0	1
CO 4	2	1	1	0	0	0	2	2	0	0	0	0	1
CO 5	2	0	0	0	0	0	2	2	0	0	0	0	1
CO 6	2	0	0	0	0	0	2	2	0	0	0	0	1

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6	✓	✓		✓